



RV College of Engineering®
(An autonomous institution affiliated to VTU, Belagavi)

Department of Master of Computer Applications

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IV SEMESTER – MCA492L Technical Seminar

Security, Fairness, and Ethical Governance in AI-Native OS Process Management

As Artificial Intelligence (AI) workloads become deeply embedded within operating system (OS) kernels through AI-Native Operating Systems (AI-OS), a new class of systemic risks has emerged extending beyond conventional cybersecurity into algorithmic fairness and ethical governance. Unlike traditional OS schedulers that allocate resources through deterministic rule-based policies, AI-driven schedulers introduce probabilistic, learned decision-making at the kernel level — decisions that can exhibit bias, be exploited by adversarial workloads, and operate with little transparency or auditability. In multi-tenant cloud and edge computing environments, inequitable resource allocation can cause systematic disadvantage to lower-priority processes, latency spikes in critical inference pipelines, and vulnerabilities to scheduler-targeted attacks. The integration of ethical AI principles — transparency, accountability, and fairness — into OS-level process management is therefore a practical necessity for deploying trustworthy computing infrastructures across healthcare, finance, and autonomous systems. This seminar will investigate the security vulnerabilities, fairness challenges, and ethical governance frameworks applicable to AI-driven scheduling and process management in AI-native operating systems.

Research into security and fairness within OS scheduling has historically been treated as separate concerns with limited cross-disciplinary synthesis at the kernel level. On the security front, studies have identified scheduler-based side-channel attacks — including cache-timing and shared-resource inference attacks — wherein adversarial processes manipulate scheduling queues to extract information about co-located workloads. Frameworks such as Intel TDX and AMD SEV provide process-level memory isolation, yet were not designed for AI-driven schedulers and offer incomplete protection against adversarial workload injection. On the fairness side, classical metrics including Max-Min Fairness and Jain's Fairness Index have been applied to CPU and network scheduling, but their adaptation to heterogeneous AI workloads involving GPU/TPU resources remains an open challenge. Dominant Resource Fairness (DRF) offers a more comprehensive multi-resource framework, though its computational overhead limits real-time applicability at the kernel level. Governance frameworks such as the European Union (EU) AI Act and IEEE's Ethically Aligned Design provide high-level principles, but their operationalization within OS process management lacks concrete technical specification, representing a significant research gap.

This seminar will conduct a systematic literature survey and conceptual analysis of security, fairness, and ethical governance in AI-native OS process management. It will examine known attack vectors targeting AI-driven schedulers, evaluate secure scheduling frameworks and their limitations, survey fairness metrics for heterogeneous AI resource allocation, and assess how governance frameworks can be operationalized as OS design requirements—including explainable scheduling logs and kernel-level audit trails. The anticipated outcomes are: (1) a characterization of the security threat landscape for AI-OS schedulers covering adversarial injection, side-channel exploitation, and privilege escalation; (2) a comparative analysis of fairness metrics (Max-Min Fairness, Jain's Index, DRF) for heterogeneous GPU/TPU environments; and (3) actionable design principles for auditable, ethically governed AI-native process management aligned with the EU AI Act and IEEE's Ethically Aligned Design. These outcomes aim to bridge the gap between high-level AI governance policy and its concrete OS-level implementation, informing future research in trustworthy AI-native operating systems.

Signatures

Student

Internal Guide

**Committee Member/
External Evaluator**

Director